

Microeconomics Revision Essay (7) Perfect Competition and Monopoly

- (a) Explain why perfect competition might be expected to result in an allocation of resources which is both productively and allocatively efficient. (20 marks)
- (b) If a market is dominated by a few large firms, the government should take action to increase the number of competitors. Discuss. (30 marks)

Outline the characteristics of a perfectly competitive market

Price Taking Firms: Each firm produces only a small percentage of total market output. It therefore exercises no control over the market price. For example it cannot restrict output in the hope of forcing up the existing market price. No individual buyer has any control over the market. I.e. there is no monopsony power. It follows from these two assumptions that buyers and sellers must regard the market price as beyond their control

Freedom of Entry and Exit: There is perfect freedom of entry and exit from the industry. Firms face few or no sunk costs that might impede movement in and out of the market. This ensures all firms only make normal profits in the long run

Homogeneous Products: Firms in the market produce homogeneous products that are perfect substitutes for each other. This leads to each firms being a price taker in the market and facing a perfectly elastic demand curve (average revenue curve) for their product

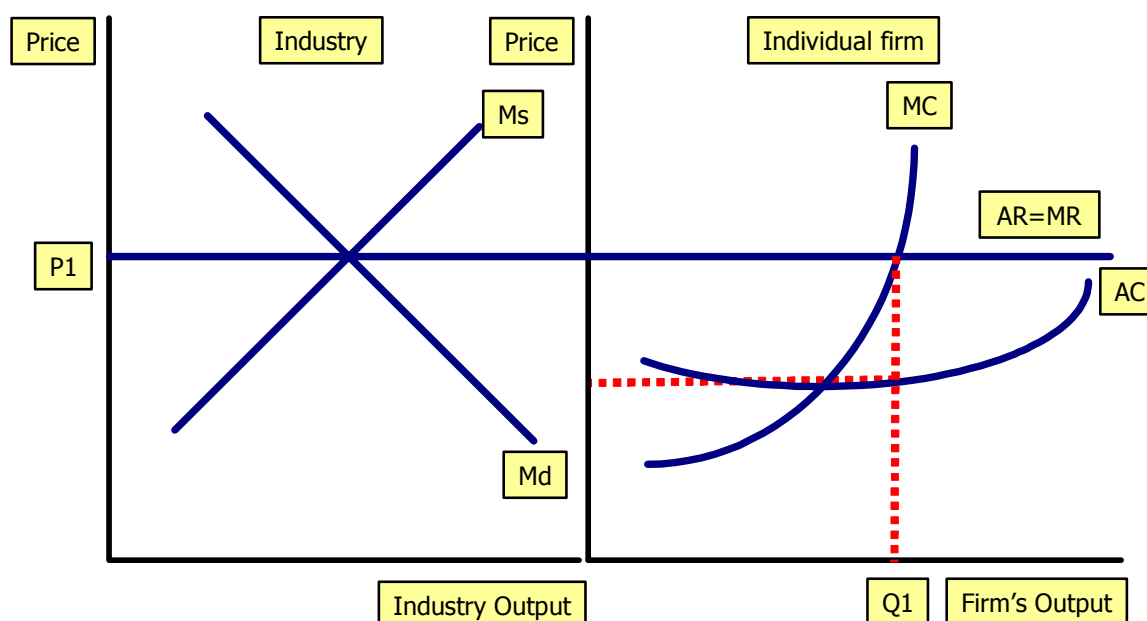
Perfect knowledge - consumers have perfect information about prices and products.

Absence of Externalities: There are no externalities arising from production or consumption

Short Run Price and Output under Perfect Competition

All individual firms are price takers. This means they face a perfectly elastic demand curve. If any firm raises its price demand will fall to zero as consumers, with perfect knowledge, switch to other producers.

Draw the diagram showing short run equilibrium for the market and an individual firm

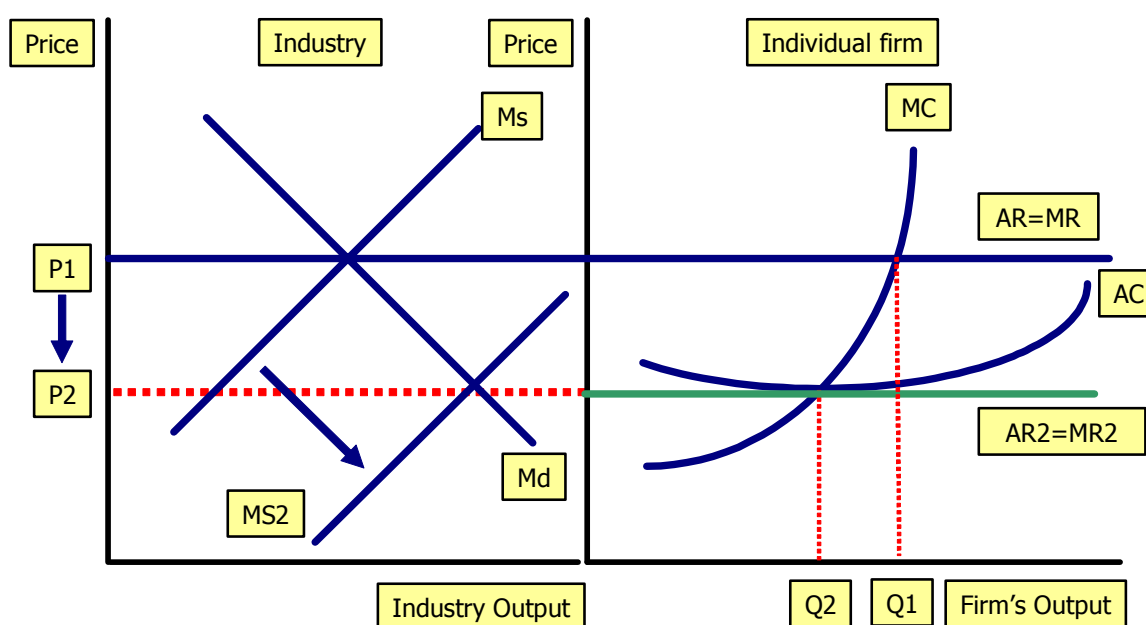


The interaction of market demand and market supply determines the equilibrium “market-clearing” price for the industry. In the diagram below, a market price P_1 is established and output Q_1 is produced. This price is taken by each of the firms. For the firm the profit maximising output is at Q_1 where $MC=MR$.

Since total revenue exceeds total cost, the firm is making abnormal (supernormal) profits. This is not necessarily the case for all firms. It depends on their short run cost curves. Some firms may be experiencing sub-normal profits if average cost exceeds the ruling market price.

Long Run Price and Output under Perfect Competition

If most firms are making abnormal profits in the short run, this encourages the entry of new firms into the industry, causing an outward shift in the market supply curve from S to S_2 , and forcing down the market price towards P_2 . The new profit-maximising output at price P_2 is output Q_2 .



The increase in supply has reduced the market price until price = long run average cost. At this point each firm is making normal profits only. There is no further incentive for movement of firms in and out of the industry and a long-run equilibrium has been established.

Explain why the long run equilibrium leads to both allocative and productive efficiency

Allocative efficiency occurs when the value consumers place on a good (reflected in the price people are willing and able to pay) equals the cost of the resources used up in production. The condition required is that price = marginal cost.

Pareto defined allocative efficiency as where no one could be made better off without making someone else at least as worse off. This can be illustrated using a production possibility frontier - all points that lie on the PPF are allocatively efficient because we cannot produce more of one product without affecting the amount of all other products available.

Productive efficiency relates to the efficiency with which factor inputs are used in producing goods and services. It is achieved when the output is produced at minimum average total cost (AC).

Perfect Competition and Efficiency

Perfect competition displays high levels of economic efficiency. In both the short and long run, price is equal to marginal cost ($P=MC$) and allocative efficiency is achieved.

Productive efficiency occurs when price is equal to average cost at its minimum point. This is not achieved in the short run, but is attained in the long run. The long run of perfect competition, therefore, exhibits optimal levels of economic efficiency.

The key assumption that gives this result is that of **free entry and exit** into and out of the industry. When supernormal profits are being made, market supply increases through the entry of new suppliers. When the market/industry is making sub-normal profits (economic losses), there will be a net loss of capital from the market (raising price and restoring profits towards their normal level).

(B) If a market is dominated by a few large firms, discuss whether the government should take action to increase the number of competitors (30 marks)

Market concentration

“A market dominated by a few large firms” - suggests an industry operating either as an oligopoly, a duopoly or a monopoly - where market concentration is high and where existing (incumbent) firms have power in setting prices, advertising spend, research and development etc.

Cost-benefit analysis

The discussion in part (B) should focus on the costs and benefits of imperfectly competitive markets and whether there is a case for government intervention to increase market contestability - justified on the basis of perceived market failure and loss of economic and social welfare

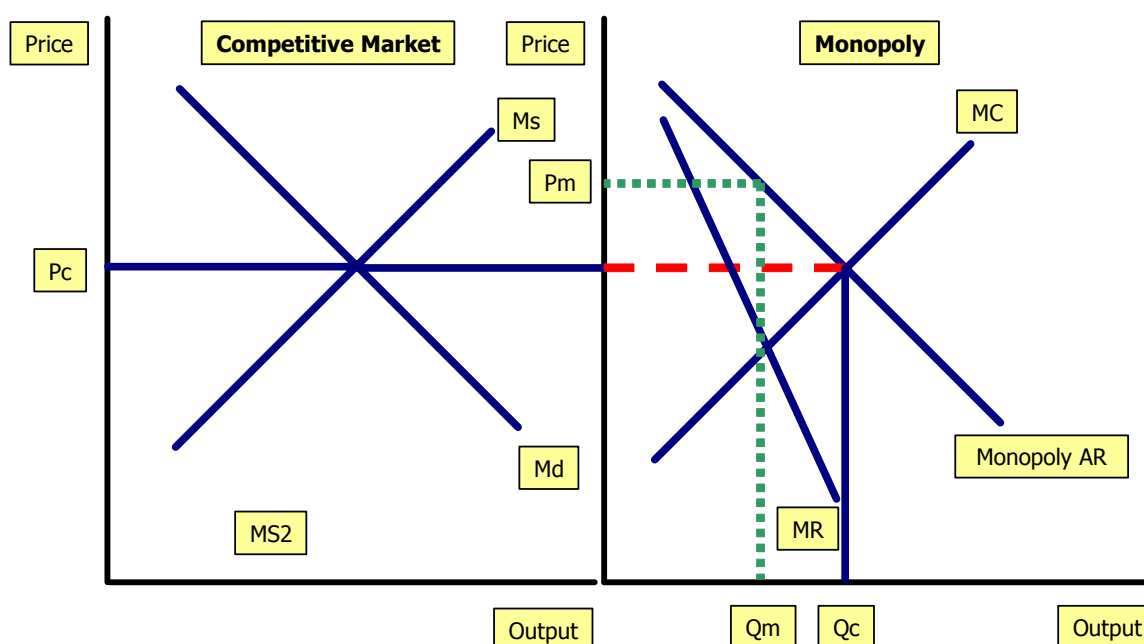
Use real world examples to develop your answer

The examiners will always reward references to examples of relevant industries and markets where this issue is of current importance (good examples to use might include telecommunications, financial services, the major domestic utilities and postal services etc.)

The Economic Case for Government Intervention

(1) Monopoly power leads to a loss of economic efficiency

Monopolists can earn abnormal profits in the short and long run at the expense of economic efficiency. The price charged by a profit-maximising monopolist is higher than average cost and neither allocative nor productive efficiency is achieved. This can be shown by the deadweight loss of consumer welfare (the social cost of monopoly). Compared to a competitive industry and assuming the same cost structure, price will be higher and output will be lower under monopoly (shown in the diagram below)



(2) Firms with market power can engage in anti-competitive behaviour

Firms within an oligopoly may decide to recognise their interdependence and move away from aggressive price competition through implicit and explicit forms of price collusion. Cartel behaviour also leads to a loss of allocative efficiency as prices remain higher than they would under normal market conditions.

(3) Markets that lack contestability fail to produce the benefits leading to dynamic efficiency

The dynamic benefits that arise from genuine competition within markets may be lost if industries become highly concentrated.

Firstly, competitive disciplines 'sorts' firms - it forces under-performing firms out of the market and it gradually shifts market share to more cost and price efficient firms and those suppliers who have the best non-price competitiveness.

Competition also encourages firms to innovate and adopt best-practise techniques (also known as benchmarking) - particularly if they want to succeed in highly competitive global markets.

Supporters of market liberalisation claim that there are substantial gains in economic welfare to be achieved from increasing the contestability of many markets and industries. They point to the success of liberalisation in telecommunications; the gas and electricity supply markets and in several other industries now open to genuine competition.

Should the Government Intervene To Increase The Number of Competitors?

The issue is fairly specific in part (B) - intervention should take the form of measures designed to open up markets to increased competition - a process known as market liberalisation or deregulation

The Case for Monopoly / Alternative Forms of Intervention / No Intervention

Scale Economies and the Natural Monopoly Argument

Firms with monopoly power (high market share) might be better able to exploit economies of scale - leading to gains in productive efficiency. Some industries may be classified as natural monopolies and

suitable more to regulation (e.g. through price capping) rather than a deliberate attempt to allow the entry of new suppliers.

A natural monopoly occurs when total market demand is most economically and efficiently satisfied by a single producer and where competition results in duplication and wasted capital investment.

Research and Development and Dynamic Efficiency

As firms are able to earn abnormal profits in the long run there may be a faster rate of technological development that will reduce costs and produce better quality products for consumers. This is because the monopolist will invest profits (producer surplus) into research and development.

Benefits from Price Discrimination and Cross Subsidisation

Firms with market power have the opportunity to engage in price discrimination. Although the main aim of this is to increase economic profits, some consumers do stand to gain from cross-subsidisation of loss-making activities financed from profits made in other parts of a firm's major markets. For example, the provision of a universal postal system in Britain might be threatened if new suppliers are allowed to enter the market.

Alternative Forms of Regulation

Market liberalisation is not the only option. Even where monopolies persist, the government (or its appointed agencies) can regulate through price capping and other forms of intervention. For example, the Competition Commission has the statutory power to fine businesses found guilty of anti-competitive practices (fines can be up to 10% of annual turnover) and the Office of Fair Trading also investigates markets to ensure that consumer welfare is not damaged.

A tough regulatory regime can bring some of the benefits of the disciplines of competition without sacrificing some of the cost advantages of large-scale production and market power.

Markets are becoming more Contestable anyway - no need for specific Government Intervention

An alternative line of argument is that advances in technology are changing the cost structures of many industries and making them more contestable, independent of direct government intervention. Technological progress has brought down the entry costs in some markets (an increase in capital mobility) E.g. Desktop publishing for magazines and the rapid expansion of E-commerce which has seen the emergence of new players in the travel sector, online bookselling and banking/financial services - in many cases challenging the dominance of the established market players

In an essay such as this, it is virtually impossible to come to a definitive conclusion. The case for government intervention to liberalise markets might have to be made on a case-by-case basis depending on the individual structure - conduct and performance from a given industry.